# Design an Embedded C Firmware for Fault Detection Using Threshold Logic in Industrial Systems

## Objective

Design a simple Embedded C program that detects faults in an industrial system by comparing a sensor value against predefined threshold limits.

## Algorithm

1. Start the program.
2. Read the sensor value.
3. Compare the value with the threshold range (50–200).
4. If the value is below 50, report a fault.
5. If the value is above 200, report a fault.
6. Otherwise, report normal operation.
7. Stop.

## Embedded C Program

```c
#include <stdio.h>

int main() {
    int sensor;

    printf("Industrial Fault Detection System\n");
    printf("Threshold Range: 50 - 200\n\n");
    printf("Enter sensor value: ");
    scanf("%d", &sensor);

    if(sensor < 50){
      printf("Status: FAULT\n");
      printf("Reason: Value below minimum threshold.\n");
    }
    else if(sensor > 200){
      printf("Status: FAULT\n");
      printf("Reason: Value above maximum threshold.\n");
    }
    else{
      printf("Status: NORMAL\n");
      printf("Sensor value is within the safe range.\n");
    }

    return 0;
}
```

## Program Screenshots

main.c

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Run

```c
#include <stdio.h>

int main() {
    int sensor;

    printf("Industrial Fault Detection System\n");
    printf("Threshold Range: 50 - 200\n\n");

    printf("Enter sensor value: ");
    scanf("%d", &sensor);

    if (sensor < 50) {
        printf("Status: FAULT\n");
        printf("Reason: Value below minimum threshold.\n");
    }
    else if (sensor > 200) {
        printf("Status: FAULT\n");
        printf("Reason: Value above maximum threshold.\n");
    }
    else {
        printf("Status: NORMAL\n");
        printf("Sensor value is within the safe range.\n");
    }

    return 0;
}
```

Output

Clear

```
Industrial Fault Detection System
Threshold Range: 50 - 200

Enter sensor value: 120
Status: NORMAL
Sensor value is within the safe range.


=== Code Execution Successful ===
```

Output

Clear

```
Industrial Fault Detection System
Threshold Range: 50 - 200

Enter sensor value: 120
Status: NORMAL
Sensor value is within the safe range.


=== Code Execution Successful ===
```

Output

Clear

```
Industrial Fault Detection System
Threshold Range: 50 - 200

Enter sensor value: 250
Status: FAULT
Reason: Value above maximum threshold.


=== Code Execution Successful ===
```

## Conclusion

The firmware implements threshold-based fault detection. Values outside the safe range (50–200) are classified as faults, while values within the range indicate normal operation. This logic is commonly used in industrial monitoring systems for temperature, pressure, and process control.